Лабораторная работа №2

Моделирование сетей передачи данных

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Цель лабораторной работы

- Познакомиться с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получить навыки проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

Установка необходимого программного обеспечения

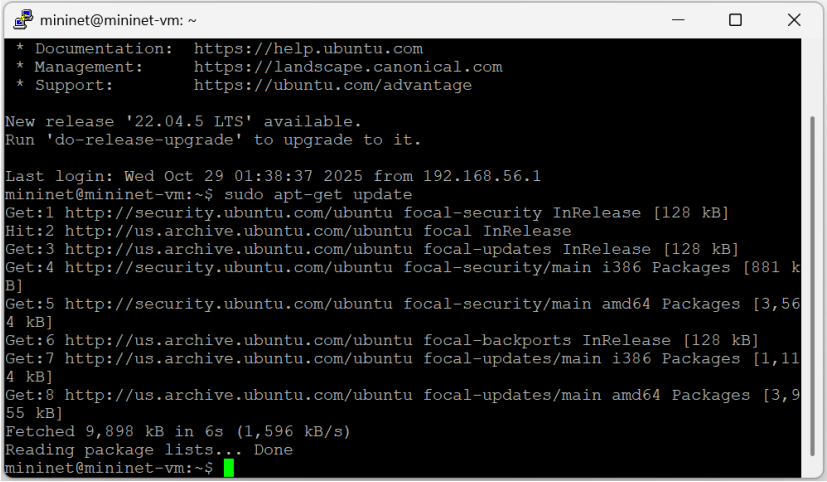

Figure 1: Обновление репозиториев программного обеспечения

Установка необходимого программного обеспечения

```
mininet@mininet-vm:~$ sudo apt-get install iperf3
Reading package lists... Done
Building dependency tree
Reading state information... Done
iperf3 is already the newest version (3.7-3).
0 upgraded, 0 newly installed, 0 to remove and 394 not upgraded.
mininet@mininet-vm:~$
```

Figure 2: Установка iperf3

Установка необходимого программного обеспечения

```
mininet@mininet-vm:~$ sudo apt-get install git jq gnuplot-nox evince
Reading package lists... Done
Building dependency tree
Reading state information... Done
gnuplot-nox is already the newest version (5.2.8+dfsg1-2).
evince is already the newest version (3.36.10-0ubuntu1).
git is already the newest version (1:2.25.1-1ubuntu3.14).
jq is already the newest version (1.6-1ubuntu0.20.04.1).
0 upgraded, 0 newly installed, 0 to remove and 394 not upgraded.
mininet@mininet-vm:~$
```

Figure 3: Установка необходимого дополнительного программного обеспечения на виртуальную машину

Установка необходимого программного обеспечения

```
mininet@mininet-vm:~$ cd /tmp/
mininet@mininet-vm:/tmp$ git clone https://github.com/ekfoury/iperf3_plotter.git
Cloning into 'iperf3_plotter'...
remote: Enumerating objects: 74, done.
remote: Total 74 (delta 0), reused 0 (delta 0), pack-reused 74 (from 1)
Unpacking objects: 100% (74/74), 100.09 KiB | 915.00 KiB/s, done.
mininet@mininet-vm:/tmp$ cd /tmp/iperf3_plotter
mininet@mininet-vm:/tmp/iperf3_plotter$ sudo cp plot_* /usr/bin
mininet@mininet-vm:/tmp/iperf3_plotter$ sudo cp *.sh /usr/bin
mininet@mininet-vm:/tmp/iperf3_plotter$
```

Figure 4: Развертывание iperf3_plotter

Интерактивные эксперименты

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  cd6aef0df44e2ea7e1dd4c6d65aefa53
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth add mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  cd6aef0df44
e2ea7e1dd4c6d65aefa53
root@mininet-vm:~# xauth list $DISPLAY
mininet-vm/unix:10  MIT-MAGIC-COOKIE-1  cd6aef0df44e2ea7e1dd4c6d65aefa53
root@mininet-vm:~#
```

Figure 5: Исправление прав запуска X-соединения

Интерактивные эксперименты

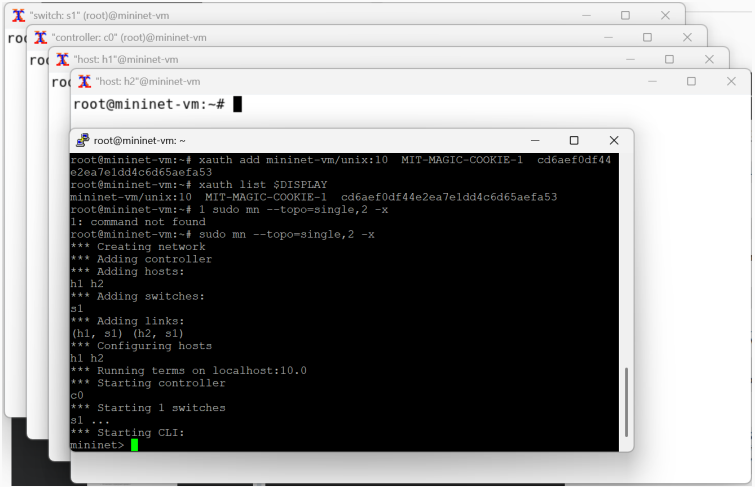

Figure 6: Создание простейшей топологии, состоящей из двух хостов и коммутатора

Интерактивные эксперименты

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo:  s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=1202>
<Host h2: h2-eth0:10.0.0.2 pid=1206>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=1211>
<Controller c0: 127.0.0.1:6653 pid=1195>
mininet>
```

Figure 7: Просмотр параметров топологии

Интерактивные эксперименты

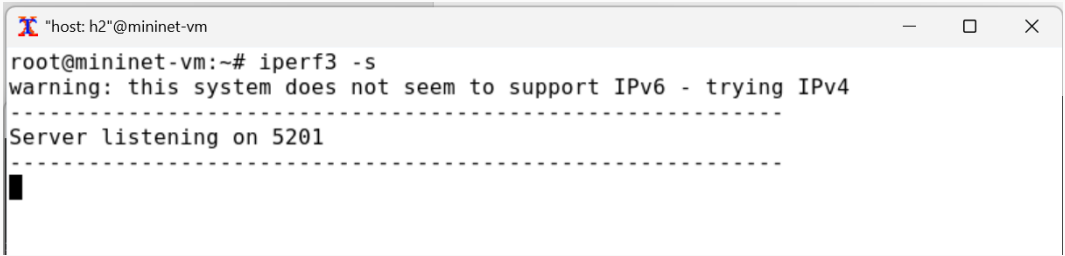

Figure 8: Запуск сервера iperf3 в терминале h2

Интерактивные эксперименты

"host: h1"@mininet-vm

```
root@mininet-vm:~# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 58386 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  1.81 GBytes  15.5 Gbits/sec    0   8.35 MBytes
[  7]   1.00-2.00   sec  1.89 GBytes  16.3 Gbits/sec    0   8.35 MBytes
[  7]   2.00-3.00   sec  1.85 GBytes  15.9 Gbits/sec    0   8.35 MBytes
[  7]   3.00-4.01   sec  1.72 GBytes  14.8 Gbits/sec    0   8.35 MBytes
[  7]   4.01-5.00   sec  1.75 GBytes  15.1 Gbits/sec    0   8.35 MBytes
[  7]   5.00-6.00   sec  1.02 GBytes  8.80 Gbits/sec    0   8.35 MBytes
[  7]   6.00-7.00   sec   914 MBytes  7.68 Gbits/sec    0   8.35 MBytes
[  7]   7.00-8.00   sec   986 MBytes  8.26 Gbits/sec    0   8.35 MBytes
[  7]   8.00-9.00   sec   949 MBytes  7.97 Gbits/sec    0   8.35 MBytes
[  7]   9.00-10.00  sec  1.01 GBytes  8.64 Gbits/sec    0   8.35 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  13.8 GBytes  11.9 Gbits/sec    0             sender
[  7]   0.00-10.00  sec  13.8 GBytes  11.9 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:~#
```

Figure 9: Запуск клиента iperf3 в терминале хоста h1

Интерактивные эксперименты

"host: h2"@mininet-vm

```
root@mininet-vm:~# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 10.0.0.1, port 58384
[  7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 58386
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-1.00   sec  1.81 GBytes  15.5 Gbits/sec
[  7]   1.00-2.00   sec  1.89 GBytes  16.1 Gbits/sec
[  7]   2.00-3.00   sec  1.86 GBytes  16.0 Gbits/sec
[  7]   3.00-4.00   sec  1.72 GBytes  14.7 Gbits/sec
[  7]   4.00-5.00   sec  1.75 GBytes  15.0 Gbits/sec
[  7]   5.00-6.00   sec  1.03 GBytes  8.90 Gbits/sec
[  7]   6.00-7.01   sec   905 MBytes  7.53 Gbits/sec
[  7]   7.01-8.00   sec   996 MBytes  8.41 Gbits/sec
[  7]   8.00-9.01   sec   949 MBytes  7.87 Gbits/sec
[  7]   9.01-10.00  sec  1.01 GBytes  8.75 Gbits/sec
[  7]  10.00-10.00  sec   704 KBytes  1.46 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-10.00  sec  13.8 GBytes  11.9 Gbits/sec                  receiver
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Figure 10: Остановка сервера iPerf3 в терминале хоста h2

Интерактивные эксперименты

```
mininet> h2 iperf3 -s &
mininet> h1 iperf -c h2
connect failed: Connection refused
mininet> h1 iperf3 -c h2
Connecting to host 10.0.0.2, port 5201
[  5] local 10.0.0.1 port 58416 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  5]   0.00-1.00   sec  1.65 GBytes  14.2 Gbits/sec    0   8.37 MBytes
[  5]   1.00-2.00   sec  1.63 GBytes  13.9 Gbits/sec    0   8.37 MBytes
[  5]   2.00-3.00   sec  1.61 GBytes  13.9 Gbits/sec    0   8.37 MBytes
[  5]   3.00-4.00   sec   978 MBytes  8.20 Gbits/sec    0   8.37 MBytes
[  5]   4.00-5.01   sec   801 MBytes  6.65 Gbits/sec    0   8.37 MBytes
[  5]   5.01-6.00   sec   849 MBytes  7.19 Gbits/sec    0   8.37 MBytes
[  5]   6.00-7.01   sec   818 MBytes  6.82 Gbits/sec    0   8.37 MBytes
[  5]   7.01-8.00   sec   981 MBytes  8.28 Gbits/sec    0   8.37 MBytes
[  5]   8.00-9.00   sec  1.43 GBytes  12.3 Gbits/sec    0   8.37 MBytes
[  5]   9.00-10.00  sec   889 MBytes  7.43 Gbits/sec    0   8.37 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec  11.5 GBytes  9.89 Gbits/sec    0             sender
[  5]   0.00-10.01  sec  11.5 GBytes  9.88 Gbits/sec                  receiver

iperf Done.
mininet> h2 killall iperf3
warning: this system does not seem to support IPv6 - trying IPv4
iperf3: error - unable to start listener for connections: Address already in use
iperf3: exiting
mininet> h2 killall iperf3
iperf3: no process found
mininet>
Interrupt
mininet> h2 killall -9 iperf3
iperf3: no process found
mininet> h2 iperf3 -s &
mininet>
Interrupt
mininet>
```

Figure 11: Запуск сервера iperf3 на хосте h2, запуск клиента iperf3 на хосте h1, остановка серверного процесса

Интерактивные эксперименты

"host: h2"@mininet-vm

```
Server listening on 5201
-----------------------------------------------------------
Killed
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Figure 12: Запуск сервера iperf3 в терминале h2

Интерактивные эксперименты

"host: h1"@mininet-vm

```
[  7]   8.00-9.01   sec  1.00 GBytes  8.60 Gbits/sec    1   8.04 MBytes
[  7]   9.01-10.00  sec  1.38 GBytes  11.9 Gbits/sec    0   8.04 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  13.0 GBytes  11.2 Gbits/sec    2             sender
[  7]   0.00-10.00  sec  13.0 GBytes  11.2 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:~# iperf3 -c 10.0.0.2 -t 5
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 58420 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.01   sec  1.78 GBytes  15.1 Gbits/sec    0   8.04 MBytes
[  7]   1.01-2.00   sec  1.84 GBytes  15.9 Gbits/sec    0   8.04 MBytes
[  7]   2.00-3.01   sec  1.76 GBytes  15.1 Gbits/sec    0   8.04 MBytes
[  7]   3.01-4.00   sec  1005 MBytes  8.45 Gbits/sec    1   8.04 MBytes
[  7]   4.00-5.01   sec   946 MBytes  7.88 Gbits/sec    0   8.04 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-5.01   sec  7.28 GBytes  12.5 Gbits/sec    1             sender
[  7]   0.00-5.01   sec  7.28 GBytes  12.5 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:~#
```

Интерактивные эксперименты

"host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -s -i 2
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Figure 14: Запуск сервера iperf3 в терминале h2 с 2-секундным интервалом времени отсчёта

"host: h1"@mininet-vm

```
iperf Done.
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -i 2
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 58540 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-2.00   sec  1.50 GBytes  6.44 Gbits/sec    0   8.09 MBytes
[  7]   2.00-4.00   sec  1.64 GBytes  7.07 Gbits/sec    1   8.09 MBytes
[  7]   4.00-6.01   sec  1.87 GBytes  8.00 Gbits/sec    0   8.09 MBytes
[  7]   6.01-8.00   sec  1.72 GBytes  7.40 Gbits/sec    0   8.09 MBytes
[  7]   8.00-10.00  sec  1.69 GBytes  7.26 Gbits/sec    0   8.09 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  8.42 GBytes  7.23 Gbits/sec    1             sender
[  7]   0.00-10.02  sec  8.42 GBytes  7.22 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 15: Запуск клиента iperf3 в терминале h1 с 2-секундным интервалом времени отсчёта

Интерактивные эксперименты

Figure 16: Запуск сервера iperf3 в терминале h2

Интерактивные эксперименты

"host: h1"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -n 16G
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 58544 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.01   sec   870 MBytes  7.24 Gbits/sec    1   8.24 MBytes
[  7]   1.01-2.01   sec   749 MBytes  6.29 Gbits/sec    0   8.24 MBytes
[  7]   2.01-3.00   sec   810 MBytes  6.83 Gbits/sec    0   8.24 MBytes
[  7]   3.00-4.00   sec   789 MBytes  6.63 Gbits/sec    0   8.24 MBytes
[  7]   4.00-5.00   sec   972 MBytes  8.15 Gbits/sec    0   8.24 MBytes
[  7]   5.00-6.00   sec   785 MBytes  6.58 Gbits/sec    1   8.24 MBytes
[  7]   6.00-7.00   sec   795 MBytes  6.67 Gbits/sec    1   8.24 MBytes
[  7]   7.00-8.00   sec   702 MBytes  5.88 Gbits/sec    0   8.24 MBytes
[  7]   8.00-9.00   sec  1.13 GBytes  9.71 Gbits/sec    0   8.24 MBytes
[  7]   9.00-10.00  sec   808 MBytes  6.76 Gbits/sec    1   8.24 MBytes
[  7]  10.00-11.00  sec   801 MBytes  6.70 Gbits/sec    2   8.24 MBytes
[  7]  11.00-12.00  sec   795 MBytes  6.70 Gbits/sec    0   8.24 MBytes
[  7]  12.00-13.00  sec   861 MBytes  7.20 Gbits/sec    0   8.24 MBytes
[  7]  13.00-14.00  sec   886 MBytes  7.46 Gbits/sec    0   8.24 MBytes
[  7]  14.00-15.01  sec   746 MBytes  6.20 Gbits/sec    0   8.24 MBytes
[  7]  15.01-16.00  sec   782 MBytes  6.61 Gbits/sec    0   8.24 MBytes
[  7]  16.00-17.01  sec   898 MBytes  7.49 Gbits/sec    0   8.24 MBytes
[  7]  17.01-18.00  sec   882 MBytes  7.44 Gbits/sec    0   8.24 MBytes
[  7]  18.00-19.00  sec   925 MBytes  7.77 Gbits/sec    0   8.24 MBytes
[  7]  19.00-19.46  sec   374 MBytes  6.79 Gbits/sec    0   8.24 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-19.46  sec  16.0 GBytes  7.06 Gbits/sec    6             sender
[  7]   0.00-19.49  sec  16.0 GBytes  7.05 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 17: Запуск клиента iperf3 в терминале h1 с объёмом данных 16 Гбайт

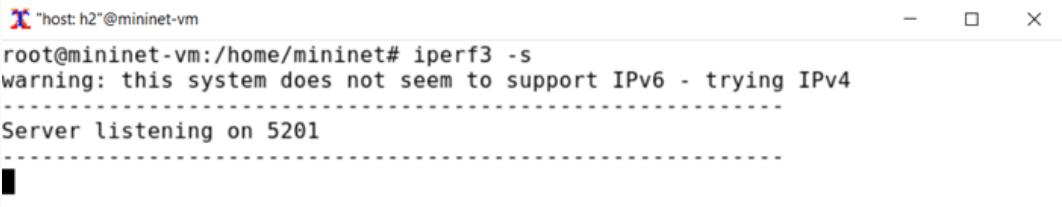

Figure 18: Запуск сервера iperf3 в терминале h2

Интерактивные эксперименты

"host: h1"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -u
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 34762 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  7]   0.00-1.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   1.00-2.00   sec   127 KBytes  1.04 Mbits/sec  90
[  7]   2.00-3.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   3.00-4.00   sec   127 KBytes  1.04 Mbits/sec  90
[  7]   4.00-5.01   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   5.01-6.00   sec   127 KBytes  1.05 Mbits/sec  90
[  7]   6.00-7.00   sec   129 KBytes  1.05 Mbits/sec  91
[  7]   7.00-8.00   sec   127 KBytes  1.04 Mbits/sec  90
[  7]   8.00-9.00   sec   129 KBytes  1.06 Mbits/sec  91
[  7]   9.00-10.00  sec   127 KBytes  1.04 Mbits/sec  90
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datag
rams
[  7]   0.00-10.00  sec  1.25 MBytes  1.05 Mbits/sec  0.000 ms  0/905 (0%)  send
er
[  7]   0.00-10.00  sec  1.25 MBytes  1.05 Mbits/sec  0.011 ms  0/905 (0%)  rece
iver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 19: Запуск клиента iperf3 в терминале h1 с протоколом UDP

"host: h2"@mininet-vm

```
-----------------------------------------------------------

^Ciperf3: interrupt - the server has terminated
root@mininet-vm:/home/mininet# iperf3 -s -p 3250
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 3250
-----------------------------------------------------------
```

Figure 20: Запуск сервера iperf3 в терминале h2 с портом прослушивания

Интерактивные эксперименты

```
"host: h1"@mininet-vm
iperf Done.
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -p 3250
iperf3: error - unable to connect to server: Connection refused
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -p 3250
Connecting to host 10.0.0.2, port 3250
[  7] local 10.0.0.1 port 52176 connected to 10.0.0.2 port 3250
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec   916 MBytes  7.68 Gbits/sec    0   8.26 MBytes
[  7]   1.00-2.01   sec   871 MBytes  7.27 Gbits/sec    0   8.26 MBytes
[  7]   2.01-3.00   sec   826 MBytes  6.97 Gbits/sec    1   8.26 MBytes
[  7]   3.00-4.01   sec   818 MBytes  6.82 Gbits/sec    0   8.26 MBytes
[  7]   4.01-5.01   sec   772 MBytes  6.47 Gbits/sec    1   8.26 MBytes
[  7]   5.01-6.00   sec   888 MBytes  7.49 Gbits/sec    1   8.26 MBytes
[  7]   6.00-7.01   sec   990 MBytes  8.21 Gbits/sec    0   8.26 MBytes
[  7]   7.01-8.01   sec   879 MBytes  7.36 Gbits/sec    0   8.26 MBytes
[  7]   8.01-9.00   sec   954 MBytes  8.09 Gbits/sec    0   8.26 MBytes
[  7]   9.00-10.01  sec   885 MBytes  7.40 Gbits/sec    0   8.26 MBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.01  sec  8.59 GBytes  7.38 Gbits/sec    3             sender
[  7]   0.00-10.01  sec  8.59 GBytes  7.37 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:/home/mininet#
```

Figure 21: Запуск клиента iperf3 в терминале h1 с портом

"host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -s -1
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Figure 22: Запуск сервера iperf3 в терминале h2 с параметром -1 (чтобы приянть только 1 клиента)

Интерактивные эксперименты

"host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -s -1
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 10.0.0.1, port 58554
[  7] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 58556
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-1.01   sec   829 MBytes  6.91 Gbits/sec
[  7]   1.01-2.01   sec   862 MBytes  7.23 Gbits/sec
[  7]   2.01-3.00   sec   781 MBytes  6.58 Gbits/sec
[  7]   3.00-4.01   sec   949 MBytes  7.93 Gbits/sec
[  7]   4.01-5.00   sec   774 MBytes  6.53 Gbits/sec
[  7]   5.00-6.00   sec   994 MBytes  8.34 Gbits/sec
[  7]   6.00-7.00   sec   840 MBytes  7.04 Gbits/sec
[  7]   7.00-8.00   sec   858 MBytes  7.21 Gbits/sec
[  7]   8.00-9.00   sec  1.05 GBytes  8.99 Gbits/sec
[  7]   9.00-10.01  sec   918 MBytes  7.62 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate
[  7]   0.00-10.01  sec  8.67 GBytes  7.44 Gbits/sec                  receiver
root@mininet-vm:/home/mininet#
```

Figure 23: Запуск клиента iperf3 в терминале h1

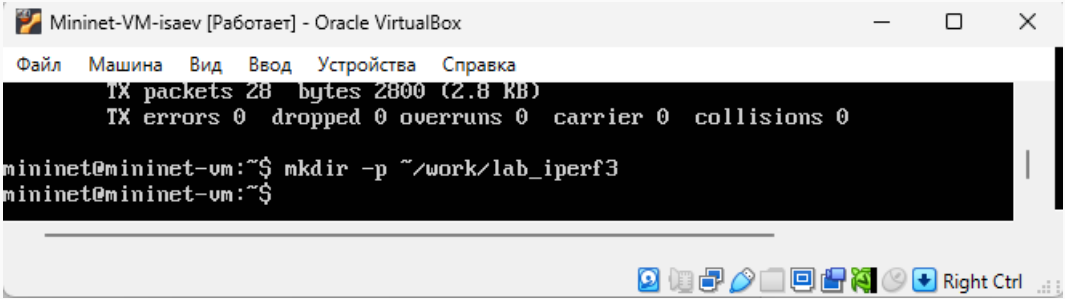

Figure 24: Создание каталога для работы над проектом

Интерактивные эксперименты

"host: h2"@mininet-vm

```
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
```

Figure 25: Запуск сервера iperf3 в терминале h2

Интерактивные эксперименты

```
"host: h1"@mininet-vm
  --username                  username for authentication
  --rsa-public-key-path       path to the RSA public key used to encrypt
                              authentication credentials

[KMG] indicates options that support a K/M/G suffix for kilo-, mega-, or giga-

iperf3 homepage at: https://software.es.net/iperf/
Report bugs to:     https://github.com/esnet/iperf
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -J
{
        "start":        {
                "connected":    [{
                                "socket":       7,
                                "local_host":   "10.0.0.1",
                                "local_port":   58560,
                                "remote_host":  "10.0.0.2",
                                "remote_port":  5201
                        }],
                "version":      "iperf 3.7",
                "system_info":  "Linux mininet-vm 5.4.0-42-generic #46-Ubuntu SM
P Fri Jul 10 00:24:02 UTC 2020 x86_64",
                "timestamp":    {
                        "time": "Wed, 29 Oct 2025 10:32:10 GMT",
                        "timesecs":     1761733930
                },
                "connecting_to":        {
                        "host": "10.0.0.2",
                        "port": 5201
                },
                "cookie":       "iym57bplp44wulc5cv3wjlmqqpy36cfehqx5",
                "tcp_mss_default":      1448,
                "sock_bufsize": 0,
                "sndbuf_actual":        87380,
                "rcvbuf_actual":        87380,
                "test_start":   {
                        "protocol":     "TCP",
                        "num_streams":  1,
                        "blksize":      131072,
                        "omit": 0,
                        "duration":     10,
```

Figure 26: Запуск клиента iperf3 в терминале h1 с параметром -J (отображение вывода в формате JSON)

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -J > /home/mininet/work/lab_i
perf3/iperf_results.json
root@mininet-vm:/home/mininet#
```

Figure 27: Экспортирование вывода результатов теста в файл

Интерактивные эксперименты

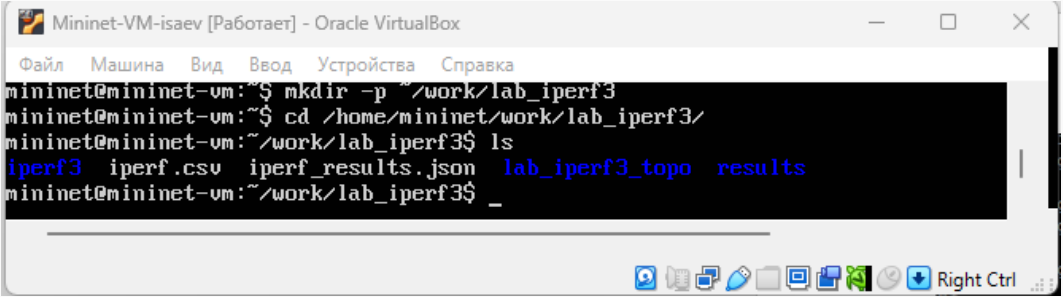

Figure 28: Проверка создания файла

Интерактивные эксперименты

```
mininet> exit
*** Stopping 1 controllers
c0
*** Stopping 8 terms
*** Stopping 2 links
..
*** Stopping 1 switches
s1
*** Stopping 2 hosts
h1 h2
*** Done
completed in 1423.250 seconds
mininet@mininet-vm:~$
```

Figure 29: Завершение работы mininet в интерактивном режиме

Интерактивные эксперименты

```
mininet@mininet-vm:~$ cd ~/work/lab_iperf3/
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
iperf3
iperf.csv
iperf_results.json
lab_iperf3_topo
results
mininet@mininet-vm:~/work/lab_iperf3$ sudo chown -R mininet:mininet ~/work
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 24
drwxrwxr-x 3 mininet mininet 4096 Oct  9 08:52 iperf3
-rw-rw-r-- 1 mininet mininet  949 Sep 27 12:14 iperf.csv
-rw-r--r-- 1 mininet mininet 7805 Oct 29 03:33 iperf_results.json
drwxrwxr-x 2 mininet mininet 4096 Oct 10 07:26 lab_iperf3_topo
drwxrwxr-x 2 mininet mininet 4096 Sep 27 12:14 results
mininet@mininet-vm:~/work/lab_iperf3$
```

Figure 30: Корректирование прав доступа к файлу JSON

Интерактивные эксперименты

```
mininet@mininet-vm: ~/work/lab_iperf3/results
-rw-rw-r-- 1 mininet mininet  949 Sep 27 12:14 iperf.csv
-rw-r--r-- 1 mininet mininet 7805 Oct 29 03:33 iperf_results.json
drwxrwxr-x 2 mininet mininet 4096 Oct 10 07:26 lab_iperf3_topo
drwxrwxr-x 2 mininet mininet 4096 Sep 27 12:14 results
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 24
drwxrwxr-x 3 mininet mininet 4096 Oct  9 08:52 iperf3
-rw-rw-r-- 1 mininet mininet  949 Sep 27 12:14 iperf.csv
-rw-r--r-- 1 mininet mininet 7805 Oct 29 03:33 iperf_results.json
drwxrwxr-x 2 mininet mininet 4096 Oct 10 07:26 lab_iperf3_topo
drwxrwxr-x 2 mininet mininet 4096 Sep 27 12:14 results
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ cd results
mininet@mininet-vm:~/work/lab_iperf3/results$ ls -l
1.dat
bytes.pdf
cwnd.pdf
MTU.pdf
retransmits.pdf
RTT.pdf
RTT_Var.pdf
throughput.pdf
mininet@mininet-vm:~/work/lab_iperf3/results$
```

Figure 31: Генерация выходных данных и последующая проверка

Вывод

- В ходе выполнения лабораторной работы познакомились с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получили навыки проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.